INDEX

Practical No.	Aim / Question	File Name
Practical No. 1	Create different blinking patterns with LEDs by connecting to GPIO pins of Raspberry Pi.	
Practical No. 2	Use Telegram to control Raspberry Pi GPIO.	2-1.pdf
Practical No. 3	Interface TM1637 with Raspberry Pi and Display current time.	
Practical No. 4	Aim is not written in the PDF.	
Practical No. 5	To replicate the signal visualization capability of an oscilloscope using Raspberry Pi and an analog-to-digital converter (ADS1115).	5_6.pdf
Practical No. 6	Interface a 16x2 LCD module with Raspberry Pi to display user-defined messages on the screen.	5_6.pdf
Practical No. 7	Interface RFID module with Raspberry Pi to read and write to RFID tags/cards.	
Practical No. 11	To interface an optical fingerprint sensor (R307/R305) via hardware UART communication (/dev/serial0) and a 16×2 character LCD using a PCF8574 (HW-61) I2C interface adapter with Raspberry Pi 4, register fingerprint biometric templates into flash memory, and perform real-time 1:N verification with status display messages.	
Practical No. 12	To interface an R307/R305 optical fingerprint sensor and a 1-channel relay module with a Raspberry Pi 4 and simulate a fingerprint-authenticated door-lock system.	
Practical No. 13	Implement basic Publisher – Subscriber using MQTT. Publish messages using paho mqtt. Subscribe and control LED.	
Practical No. 14	Design a web dashboard using Flask. Control relay module via web.	
Practical No. 17	To design and implement a Node-RED flow on Raspberry Pi that integrates a physical push-button and LED with a web dashboard. The flow must allow the users to toggle the LED from the dashboard and via the physical button.	
Practical No. 18	Write a program to read the CPU%, Memory% and CPU temperature of Raspberry Pi.	
Practical No. 19	Aim is not written in the PDF.	